

Product GRANADA

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## 1. Identification of the substance/mixture and of the company/undertaking

### 1.1 Product identifier

Trade name : GRANADA  
IFF Code : 5576693  
SDS Number : 300000559866

### 1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture : Fragrance for consumer product

### 1.3 Details of the supplier of the safety data sheet

Company : IFF (NEDERLAND) BV  
ZEVENHEUVELENWEG 60  
5048 AN TILBURG  
Netherlands  
Telephone : +31134642211  
Telefax : +31134636032  
E-mail address : sds@iff.com  
Responsible/issuing person

### 1.4 Emergency telephone number

Refer to section 16 for country specific emergency contact number.

## 2. Hazards identification

### 2.1 Classification of the substance or mixture

#### Classification (REGULATION (EC) No 1272/2008)

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| Skin irritation, Category 2          | H315: Causes skin irritation.                          |
| Eye irritation, Category 2           | H319: Causes serious eye irritation.                   |
| Skin sensitisation, Category 1       | H317: May cause an allergic skin reaction.             |
| Chronic aquatic toxicity, Category 2 | H411: Toxic to aquatic life with long lasting effects. |

### 2.2 Label elements

#### Labelling (REGULATION (EC) No 1272/2008)

Hazard pictograms :



Signal word : Warning

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|                          |                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hazard statements        | : H315<br>H317<br>H319<br>H411                                                                                                | Causes skin irritation.<br>May cause an allergic skin reaction.<br>Causes serious eye irritation.<br>Toxic to aquatic life with long lasting effects.                                                                                                                                                                                                      |
| Precautionary statements | : <b>Prevention:</b><br>P261<br><br>P273<br>P280<br><br><b>Response:</b><br>P333 + P313<br><br>P337 + P313<br><br>P362 + P364 | Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray.<br>Avoid release to the environment.<br>Wear protective gloves/ eye protection/ face protection.<br><br>If skin irritation or rash occurs: Get medical advice/ attention.<br>If eye irritation persists: Get medical advice/ attention.<br>Take off contaminated clothing and wash it before reuse. |

Hazardous components which must be listed on the label:

- 118-58-1 Benzyl Salicylate
- 33704-61-9 Dihydro pentamethylindanone
- 120-57-0 HELIOTROPINE
- 80-54-6 BUTYLPHENYL METHYLPROPIONAL
- 68155-66-8, 54464-57-2, 68155-67-9, 54464-59-4 Tetramethyl Acetyloctahydronaphthalenes
- 144020-22-4 Reaction products of acetic anhydride and 1,5,10-trimethyl-1,5,9-cyclodecatriene
- 127-51-5 Alpha-Isomethyl Ionone
- 5462-06-6 3-(p-methoxyphenyl)-2-methylpropionaldehyde
- 1205-17-0 #-methyl-1,3-benzodioxole-5-propionaldehyde
- 91-64-5 COUMARIN
- 106-22-9, 7540-51-4 Citronellol
- 138-86-3, 5989-27-5, 5989-54-8 limonene
- 105-87-3 GERANYL ACETATE
- 80-56-8 Pinene

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- 106-24-1 GERANIOL
- 13466-78-9 3,7,7-trimethylbicyclo[4.1.0]hept-3-ene
- 23696-85-7 1-(2,6,6-trimethyl-1,3-cyclohexadien-1-yl)-2-buten-1-one

## 2.3 Other hazards

None reasonably foreseeable.

## 3. Composition/information on ingredients

### 3.1 Substances

Not applicable, product is a mixture.

### 3.2 Mixtures

#### Hazardous components

|                                                                         | CAS-No.<br>EC-No.<br>Registration number       | Classification<br>(REGULATION (EC) No 1272/2008)                                        | Concentration<br>[%] |
|-------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------|
| tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans) | 63500-71-0<br>603-101-00-3<br>01-0000015458-64 | Eye Irrit.2; H319                                                                       | 10 - 20              |
| Oxacyclohexadecen-2-one                                                 | 111879-80-2<br>606-092-00-4                    | Aquatic Acute1; H400<br>Aquatic Chronic1; H410                                          | 2,5 - 10             |
| benzyl salicylate                                                       | 118-58-1<br>204-262-9<br>01-2119969442-31      | Skin Sens.1B; H317<br>Aquatic Chronic3; H412                                            | 2,5 - 10             |
| 1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran          | 1222-05-5<br>214-946-9<br>01-2119488227-29     | Aquatic Chronic1; H410<br>Aquatic Acute1; H400                                          | 2,5 - 10             |
| (Z)-3-hexenyl salicylate                                                | 65405-77-8<br>265-745-8<br>01-2119987320-37    | Aquatic Chronic1; H410<br>Aquatic Acute1; H400                                          | 2,5 - 10             |
| Dihydro pentamethylindanone                                             | 33704-61-9<br>251-649-3<br>01-2119977131-40    | Aquatic Chronic2; H411<br>Skin Sens.1B; H317<br>Eye Irrit.2; H319<br>Skin Irrit.2; H315 | 3 - 5                |
| piperonal                                                               | 120-57-0                                       | Skin Sens.1B; H317                                                                      | 0,1 - 10             |

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|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------|
|                                                                                  | 204-409-7                                                                       |                                                                                                          |            |
| 2-(4-tert-butylbenzyl)propionaldehyde                                            | 80-54-6<br>201-289-8<br>01-2119485965-18                                        | Acute Tox.4; H302<br>Skin Irrit.2; H315<br>Repr.2; H361f<br>Skin Sens.1B; H317<br>Aquatic Chronic2; H411 | 1 - 2,5    |
| Tetramethyl Acetyloctahydronaphthalenes                                          | 68155-66-8, 54464-57-2, 68155-67-9, 54464-59-4<br>915-730-3<br>01-2119489989-04 | Skin Irrit.2; H315<br>Skin Sens.1B; H317<br>Aquatic Chronic1; H410                                       | 1 - 2,5    |
| 3,7-dimethylnona-1,6-dien-3-ol                                                   | 10339-55-6<br>233-732-6                                                         | Eye Irrit.2; H319<br>Skin Irrit.2; H315                                                                  | 1 - 3      |
| ##,2,2,3-pentamethylcyclopent-3-ene-1-butanol                                    | 65113-99-7<br>265-453-0                                                         | Aquatic Chronic2; H411<br>Eye Irrit.2; H319                                                              | 1 - 2,5    |
| Reaction products of acetic anhydride and 1,5,10-trimethyl-1,5,9-cyclodecatriene | 144020-22-4<br>482-22-4<br>01-0000020172-83                                     | Skin Sens.1B; H317<br>Aquatic Chronic1; H410<br>Aquatic Acute1; H400                                     | 0,25 - 2,5 |
| 3-methyl-4-(2,6,6-trimethyl-2-cyclohexen-1-yl)-3-buten-2-one                     | 127-51-5<br>204-846-3                                                           | Skin Sens.1B; H317<br>Aquatic Chronic2; H411<br>Skin Irrit.2; H315                                       | 1 - 2,5    |
| 3-(p-methoxyphenyl)-2-methylpropionaldehyde                                      | 5462-06-6<br>226-749-5                                                          | Skin Sens.1B; H317                                                                                       | 0,1 - 10   |
| vanillin                                                                         | 121-33-5<br>204-465-2                                                           | Eye Irrit.2; H319                                                                                        | 1 - 3      |
| methyl anthranilate                                                              | 134-20-3<br>205-132-4                                                           | Eye Irrit.2; H319                                                                                        | 1 - 3      |
| #-methyl-1,3-benzodioxole-5-propionaldehyde                                      | 1205-17-0<br>214-881-6                                                          | Skin Sens.1B; H317<br>Aquatic Chronic2; H411                                                             | 0,25 - 2,5 |
| coumarin                                                                         | 91-64-5<br>202-086-7                                                            | Acute Tox.4; H302<br>Skin Sens.1B; H317                                                                  | 0,1 - 10   |
| citronellol                                                                      | 106-22-9, 7540-51-4<br>203-375-0                                                | Skin Sens.1B; H317<br>Skin Irrit.2; H315                                                                 | 0,1 - 1    |

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|----------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------|
|                                                          | 01-2119453995-23                                                        | Eye Irrit.2; H319                                                                                                                   |            |
| limonene                                                 | 138-86-3, 5989-27-5,<br>5989-54-8<br>205-341-0, 227-813-5,<br>227-815-6 | Flam. Liq.3; H226<br>Skin Sens.1B; H317<br>Asp. Tox.1; H304<br>Aquatic Chronic1; H410<br>Skin Irrit.2; H315<br>Aquatic Acute1; H400 | 0,25 - 1   |
| geranyl acetate                                          | 105-87-3<br>203-341-5                                                   | Aquatic Chronic3; H412<br>Skin Sens.1; H317<br>Skin Irrit.2; H315                                                                   | 0,25 - 1   |
| pin-2(3)-ene                                             | 80-56-8<br>201-291-9<br>01-2119519223-49                                | Flam. Liq.3; H226<br>Asp. Tox.1; H304<br>Skin Irrit.2; H315<br>Skin Sens.1B; H317<br>Aquatic Acute1; H400<br>Aquatic Chronic1; H410 | 0,25 - 1   |
| geraniol                                                 | 106-24-1<br>203-377-1<br>01-2119552430-49, 01-<br>2119560621-44         | Skin Irrit.2; H315<br>Eye Dam.1; H318<br>Skin Sens.1; H317                                                                          | 0,1 - 1    |
| 2,6-di-tert-butyl-p-cresol                               | 128-37-0<br>204-881-4<br>01-2119565113-46, 01-<br>2119555270-46         | Aquatic Chronic1; H410<br>Aquatic Acute1; H400                                                                                      | 0,1 - 0,25 |
| 3,7,7-trimethylbicyclo[4.1.0]hept-3-ene                  | 13466-78-9<br>236-719-3                                                 | Flam. Liq.3; H226<br>Asp. Tox.1; H304<br>Skin Sens.1; H317<br>Skin Irrit.2; H315                                                    | 0,1 - 1    |
| indole                                                   | 120-72-9<br>204-420-7                                                   | Acute Tox.3; H311<br>Acute Tox.4; H302                                                                                              | 0,1 - 10   |
| 1-(2,6,6-trimethyl-1,3-cyclohexadien-1-yl)-2-buten-1-one | 23696-85-7<br>245-833-2                                                 | Skin Sens.1A; H317<br>Aquatic Chronic2; H411                                                                                        | 0 - 0,1    |

For the full text of the R-phrases mentioned in this Section, see Section 16.

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## 4. First aid measures

### 4.1 Description of first aid measures

- |                         |                                                                                                                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General advice          | : Take Risk/Hazard and Safety/Precautionary phrases (section 2 or 15) into account.                                                                                                                             |
| If inhaled              | : Remove from exposure site to fresh air and keep at rest. If victim is unconscious, remove foreign bodies from the mouth. If victim has stopped breathing, give artificial respiration. Obtain medical advice. |
| In case of skin contact | : Remove contaminated clothes. Wash thoroughly with water (and soap). Contact physician if symptoms persist.                                                                                                    |
| In case of eye contact  | : Flush immediately with water for at least 15 minutes. Contact physician if symptoms persist.                                                                                                                  |
| If swallowed            | : Rinse mouth with water and obtain medical advice.                                                                                                                                                             |

### 4.2 Most important symptoms and effects, both acute and delayed

- |          |                             |
|----------|-----------------------------|
| Symptoms | : No information available. |
| Risks    | : No information available. |

### 4.3 Indication of any immediate medical attention and special treatment needed

- |           |                             |
|-----------|-----------------------------|
| Treatment | : No information available. |
|-----------|-----------------------------|

## 5. Firefighting measures

### 5.1 Extinguishing media

- |                                |                                                     |
|--------------------------------|-----------------------------------------------------|
| Suitable extinguishing media   | : Carbondioxide, dry chemical, foam.                |
| Unsuitable extinguishing media | : Do not use a direct waterjet on burning material. |

### 5.2 Special hazards arising from the substance or mixture

- |                                      |                             |
|--------------------------------------|-----------------------------|
| Specific hazards during firefighting | : Water may be ineffective. |
|--------------------------------------|-----------------------------|

### 5.3 Advice for firefighters

- |                     |                                          |
|---------------------|------------------------------------------|
| Further information | : Standard procedure for chemical fires. |
|---------------------|------------------------------------------|

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## 6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Avoid inhalation and contact with skin and eyes. A self-contained breathing apparatus is recommended in case of a major spill.

## 6.2 Environmental precautions

Environmental precautions : Keep away from drains, surface- and groundwater and soil.

## 6.3 Methods and materials for containment and cleaning up

Methods for cleaning up : Clean up spillage promptly. Remove ignition sources. Provide adequate ventilation. Avoid excessive inhalation of vapours. Gross spillages should be contained by use of sand or inert powder and disposed of according to the local regulations.

## 6.4 Reference to other sections

Prevent spreading over a wide area (e.g. by containment or oil barriers).

## 7. Handling and storage

### 7.1 Precautions for safe handling

Advice on safe handling : Avoid excessive inhalation of concentrated vapors. Follow good manufacturing practices for housekeeping and personal hygiene. Wash any exposed skin immediately after any chemical contact, before breaks and meals, and at the end of each work period. Contaminated clothing and shoes should be thoroughly cleaned before re-use.

If appropriate, procedures used during the handling of this material should also be used when cleaning equipment or removing residual chemicals from tanks or other containers, especially when steam or hot water is used, as this may increase vapor concentrations in the workplace air. Where chemicals are openly handled, access should be restricted to properly trained employees.

Keep all heated processes at the lowest necessary temperature in order to minimize emissions of volatile chemicals into the air.

Advice on protection against fire and explosion : Keep away from ignition sources and naked flame.

### 7.2 Conditions for safe storage, including any incompatibilities

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Requirements for storage areas and containers : Store in a cool, dry, ventilated area away from heat sources. Keep containers upright and tightly closed when not in use.

## 7.3 Specific end use(s)

Specific use(s) : No information available.

## 8. Exposure controls/personal protection

### 8.1 Control parameters

Contains no substances with occupational exposure limit values.

DNEL

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran : End Use: Workers  
 Exposure routes: Skin contact  
 Potential health effects: Long-term systemic effects  
 Exposure time: 8 h  
 Value: 60 mg/kg bw/day

End Use: Workers  
 Exposure routes: Inhalation  
 Potential health effects: Long-term systemic effects  
 Exposure time: 8 h  
 Value: 22 mg/m3

End Use: General population  
 Exposure routes: Inhalation  
 Potential health effects: Long-term systemic effects  
 Exposure time: 8 h

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Value: 6,5 mg/m<sup>3</sup>

End Use: General population  
 Exposure routes: Skin contact  
 Potential health effects: Long-term systemic effects  
 Exposure time: 8 h  
 Value: 36 mg/kg bw/day

End Use: General population  
 Exposure routes: Ingestion  
 Potential health effects: Long-term systemic effects  
 Exposure time: 8 h  
 Value: 3,8 mg/kg bw/day

1-(1,2,3,4,5,6,7,8-Octahydro-  
 2,3,8,8-tetramethyl-2-  
 naphthyl)ethan-1-one

: End Use: Workers  
 Exposure routes: Skin contact  
 Potential health effects: Long-term local effects  
 Exposure time: 8 h  
 Value: 0,1011 mg/cm<sup>2</sup>

End Use: Workers  
 Exposure routes: Skin contact  
 Potential health effects: Long-term systemic effects  
 Exposure time: 8 h  
 Value: 1,73 mg/kg bw/day

End Use: Workers  
 Exposure routes: Inhalation  
 Potential health effects: Long-term systemic effects  
 Exposure time: 8 h  
 Value: 1,76 mg/m<sup>3</sup>

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: Fresh water  
 Value: 0,0044 mg/l

Marine water  
 Value: 0,00044 mg/l

Fresh water sediment  
 Value: 2 mg/kg dry weight (d.w.)

Marine sediment  
 Value: 0,394 mg/kg dry weight (d.w.)

Soil  
 Value: 0,31 mg/kg dry weight (d.w.)

1-(1,2,3,4,5,6,7,8-Octahydro-  
 2,3,8,8-tetramethyl-2-  
 naphthyl)ethan-1-one

: Fresh water  
 Value: 0,0028 mg/l

Marine water  
 Value: 0,00028 mg/l

Fresh water sediment  
 Value: 3,73 mg/kg dry weight (d.w.)

Marine sediment  
 Value: 0,75 mg/kg dry weight (d.w.)

Soil  
 Value: 0,705 mg/kg dry weight (d.w.)

## 8.2 Exposure controls

### Engineering measures

Where appropriate, use closed systems to transfer and process this material.  
 If appropriate, isolate mixing rooms and other areas where this material is used or openly handled. Maintain these areas under negative air pressure relative to the rest of the plant.

### Personal protective equipment

Respiratory protection : Use local exhaust ventilation around open tanks and other open

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sources of potential exposures in order to avoid excessive inhalation, including places where this material is openly weighed or measured. In addition, use general dilution ventilation of the work area to eliminate or reduce possible worker exposures.

No respiratory protection is required during normal operations in a workplace where engineering controls such as adequate ventilation, etc. are sufficient.

If engineering controls and safe work practices are not sufficient, an approved, properly fitted respirator with organic vapor cartridges or canisters and particulate filters should be used:

- a) while engineering controls and appropriate safe work practices and/or procedures are being implemented; or
- b) during short term maintenance procedures when engineering controls are not in normal operation or are not sufficient; or
- c) if normal operational workplace vapor concentration in the air is increased due to heat ;
- d) during emergencies; or
- e) if engineering controls and operational practices are not sufficient to reduce airborne concentrations below an established occupational exposure limit.

Hand protection : Avoid skin contact. Use chemically resistant gloves.

Eye protection : Use tight-fitting goggles, face shield or safety glasses with side shields if eye contact might occur.

Hygiene measures : To the extent deemed appropriate, implement pre-placement and regularly scheduled ascertainment of symptoms and spirometry testing of lung function for workers who are regularly exposed to this material.  
 To the extent deemed appropriate, use an experienced air sampling expert to identify and measure volatile chemicals that could be present in the workplace air to determine potential exposures and to ensure the continuing effectiveness of engineering controls and operational practices to minimize exposure.

## Environmental exposure controls

General advice : Keep away from drains, surface- and groundwater and soil.

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## 9. Physical and chemical properties

### 9.1 Information on basic physical and chemical properties

|                                        |                         |
|----------------------------------------|-------------------------|
| Appearance                             | : liquid                |
| Colour                                 | : pale yellow to yellow |
| Odour                                  | : conforms to standard  |
| Odour Threshold                        | : not determined        |
| Flash point                            | : 108 °C                |
| Lower explosion limit                  | : not determined        |
| Upper explosion limit                  | : not determined        |
| Flammability (solid, gas)              | : not determined        |
| Oxidizing properties                   | : not determined        |
| Auto-ignition temperature              | : not determined        |
| pH                                     | : not determined        |
| Melting point                          | : not determined        |
| Boiling point                          | : not determined        |
| Vapour pressure                        | : 0,02 hPa Calculated   |
| Density                                | : not determined        |
| Water solubility                       | : not determined        |
| Partition coefficient: n-octanol/water | : not determined        |
| Solubility in other solvents           | : not determined        |
| Viscosity, dynamic                     | : not determined        |
| Viscosity, kinematic                   | : not determined        |
| Relative vapour density                | : not determined        |
| Evaporation rate                       | : not determined        |

### 9.2 Other information

|                  |                   |
|------------------|-------------------|
| Refractive index | : not determined  |
| Relative density | : 0,9990 - 1,0090 |

## 10. Stability and reactivity

### 10.1 Reactivity

No hazards to be specially mentioned.

### 10.2 Chemical stability

Stable under normal conditions.

### 10.3 Possibility of hazardous reactions

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Hazardous reactions : Note: Presents no significant reactivity hazard, by itself or in contact with water. Avoid contact with strong acids, alkali or oxidizing agents.

## 10.4 Conditions to avoid

Conditions to avoid : Direct sources of heat.

## 10.5 Incompatible materials

Materials to avoid : Avoid contact with strong acids, alkali or oxidizing agents.

## 10.6 Hazardous decomposition products

Hazardous decomposition products : Carbon monoxide and unidentified organic compounds may be formed during combustion.

# 11. Toxicological information

## 11.1 Information on toxicological effects

### Acute toxicity

Acute oral toxicity : Acute toxicity estimate: > 2.000 mg/kg  
 Method: Calculation method

Acute dermal toxicity : Acute toxicity estimate: > 2.000 mg/kg  
 Method: Calculation method

### Skin corrosion/irritation

benzyl salicylate : Species: Rabbit  
 Result: No skin irritation

### Serious eye damage/eye irritation

tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans) : Species: Rabbit  
 Result: Eye irritation

### Respiratory or skin sensitisation

benzyl salicylate : maximisation study  
 Species: human  
 Result: Did not cause sensitisation on laboratory animals.  
 Test substance: 30% in petrolatum

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## **Germ cell mutagenicity**

Information on ingredients with chronic toxicity hazard(s) is available upon request.

## **Carcinogenicity**

Information on ingredients with chronic toxicity hazard(s) is available upon request.

## **Reproductive toxicity**

Information on ingredients with chronic toxicity hazard(s) is available upon request.

## **Target Organ Systemic Toxicant - Single exposure**

Information on ingredients with chronic toxicity hazard(s) is available upon request.

## **Target Organ Systemic Toxicant - Repeated exposure**

Information on ingredients with chronic toxicity hazard(s) is available upon request.

## **Aspiration hazard**

No test data available. Ingredients with an aspiration hazard are identified in section 3.

## **12. Ecological information**

### **12.1 Toxicity**

Toxicity to fish

A mixture of: (E)-oxacyclohexadec-12-en-2-one and (E)-oxacyclohexadec-13-en-2-one and a) (Z)-oxacyclohexadec-(12)-en-2-one and b) (Z)  
 : LC50: 2 mg/l  
 Exposure time: 96 h  
 Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates

A mixture of: (E)-oxacyclohexadec-12-en-2-one and (E)-oxacyclohexadec-13-en-2-one and a) (Z)-oxacyclohexadec-(12)-en-2-one and b) (Z)  
 : EC50: 0,48 mg/l  
 Exposure time: 48 h  
 Method: OECD Test Guideline 202

Toxicity to algae

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A mixture of: (E)-oxacyclohexadec-12-en-2-one and (E)-oxacyclohexadec-13-en-2-one and a) (Z)-oxacyclohexadec-(12)-en-2-one and b) (Z)

: ErC50: 5 mg/l  
 Exposure time: 72 h  
 Method: OECD Test Guideline 201

EbC50: 2,4 mg/l  
 Exposure time: 72 h  
 Method: OECD Test Guideline 201

M-Factor  
 HABANOLIDE : 1

## 12.2 Persistence and degradability

Biodegradability  
 A mixture of: (E)-oxacyclohexadec-12-en-2-one and (E)-oxacyclohexadec-13-en-2-one and a) (Z)-oxacyclohexadec-(12)-en-2-one and b) (Z)

: Result: Readily biodegradable.  
 95 %  
 Method: Directive 67/548/EEC Annex V, C.4.F.

## 12.3 Bioaccumulative potential

Bioaccumulation  
 A mixture of: (E)-oxacyclohexadec-12-en-2-one and (E)-oxacyclohexadec-13-en-2-one and a) (Z)-oxacyclohexadec-(12)-en-2-one and b) (Z)

: Remarks:  
 No data available

## 12.4 Mobility in soil

Mobility : No data available

## 12.5 Results of PBT and vPvB assessment

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

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## 12.6 Other adverse effects

Additional ecological information : There is no data available for this product.

## 13. Disposal considerations

### 13.1 Waste treatment methods

Product : Dispose of according to local regulations. Avoid disposing into drainage systems and into the environment.

Contaminated packaging : Empty containers should be taken to an approved waste handling site for recycling or disposal.

## 14. Transport information

### ADR

UN number : 3082

Description of the goods : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (OXACYCLOHEXADECEN-2-ONE)

Labels : 9

Packing group : III

Environmentally hazardous : yes

### IATA

UN number : 3082

Description of the goods : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (OXACYCLOHEXADECEN-2-ONE)

Labels : 9

Packing group : III

Environmentally hazardous : yes

### IMDG

UN number : 3082

Description of the goods : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (OXACYCLOHEXADECEN-2-ONE)

Labels : 9

Packing group : III

Marine pollutant : yes (OXACYCLOHEXADECEN-2-ONE)

**Special precautions for user** : No special precautions required.

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## 15. Regulatory information

### 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Water contaminating class : WGK 2water endangering  
 (Germany)

### 15.2 Chemical safety assessment

A Chemical Safety Assessment is not required for this substance.

## 16. Other information

### Full text of H-Statements referred to under sections 2 and 3.

|       |                                                       |
|-------|-------------------------------------------------------|
| H226  | Flammable liquid and vapour.                          |
| H302  | Harmful if swallowed.                                 |
| H304  | May be fatal if swallowed and enters airways.         |
| H311  | Toxic in contact with skin.                           |
| H315  | Causes skin irritation.                               |
| H317  | May cause an allergic skin reaction.                  |
| H318  | Causes serious eye damage.                            |
| H319  | Causes serious eye irritation.                        |
| H361f | Suspected of damaging fertility.                      |
| H400  | Very toxic to aquatic life.                           |
| H410  | Very toxic to aquatic life with long lasting effects. |
| H411  | Toxic to aquatic life with long lasting effects.      |
| H412  | Harmful to aquatic life with long lasting effects.    |

### Further information

In December 2003, the National Institute for Occupational Safety and Health ("NIOSH") published an Alert on preventing lung disease in workers who use or make flavorings [NIOSH Publication Number 2004-110]. In August 2004, the United States Flavor and Extract Manufacturers Association (FEMA) issued a report entitled "Respiratory Safety in the Flavor Manufacturing Workplace".

Both of these reports provide recommendations for reducing employee exposure and for medical surveillance in the workplace. The recommendations in these reports are generally applicable to the use of any chemical in the workplace and you are strongly urged to review both of these reports.

The report published by FEMA also contains a list of "high priority" chemicals. If any of these chemicals are present in this product at a concentration  $\geq 1.0\%$  due to an intentional addition by IFF, the chemical(s) will be identified in this safety data sheet.

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According to Regulation (EC) No. 1907/2006 the information in this safety data sheet is based on the properties of the material known to IFF at the time the data sheet was issued. The safety data sheet is intended to provide information for a health and safety assessment of the material and the circumstances, under which it is packaged, stored or applied in the workplace. For such a safety assessment International Flavors & Fragrances holds no responsibility. This document is not intended for quality assurance purposes.

## Emergency telephone number

|                |                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------|
| Austria        | +43 1 406 43 43                                                                                         |
| Belgium        | +32 70 245 245                                                                                          |
| Bulgaria       | +359 2 9154 233                                                                                         |
| Croatia        | (+385 1) 2348342                                                                                        |
| Czech Republic | +420 224 919 293 / +420 224 915 402                                                                     |
| Denmark        | +45 82 12 12 12                                                                                         |
| Estonia        | 16662 (National), International (+372) 626 93 90                                                        |
| Finland        | +358 9 471977                                                                                           |
| France         | + 33 (0)1 45 42 59 59                                                                                   |
| Germany        | +31 13 4642 211                                                                                         |
| Greece         | +31 13 4642 211                                                                                         |
| Hungary        | (+36-80) 201-199                                                                                        |
| Iceland        | +354 543 2222                                                                                           |
| Ireland        | +353 1 8092566 / +353 1 8379964                                                                         |
| Italy          | +39 06 68593726                                                                                         |
| Latvia         | +371 67042473                                                                                           |
| Lithuania      | +370 5 236 20 52 or +370 687 53378                                                                      |
| Luxembourg     | +352 8002 5500                                                                                          |
| Malta          | +356 21224071                                                                                           |
| Netherlands    | +31 30 2748888 (Only for the purpose of informing medical personnel in cases of acute intoxications).   |
| Norway         | +47 22 59 13 00                                                                                         |
| Poland         | +31 13 4642 211                                                                                         |
| Portugal       | 808 250 143                                                                                             |
| Poland         | +31 13 4642 211                                                                                         |
| Portugal       | 808 250 143                                                                                             |
| Romania        | +31 13 4642 211                                                                                         |
| Slovakia       | +31 13 4642 211                                                                                         |
| Slovenia       | +31 13 4642 211                                                                                         |
| Spain          | +34 91 562 04 20 (only for the purpose of informing medical personnel in cases of acute intoxications). |

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|                |                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------|
| Sweden         | +46 112                                                                                           |
| United Kingdom | For medical professionals only +44 845 46 47 (England and Wales)<br>+ 44 8454 24 24 24 (Scotland) |

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